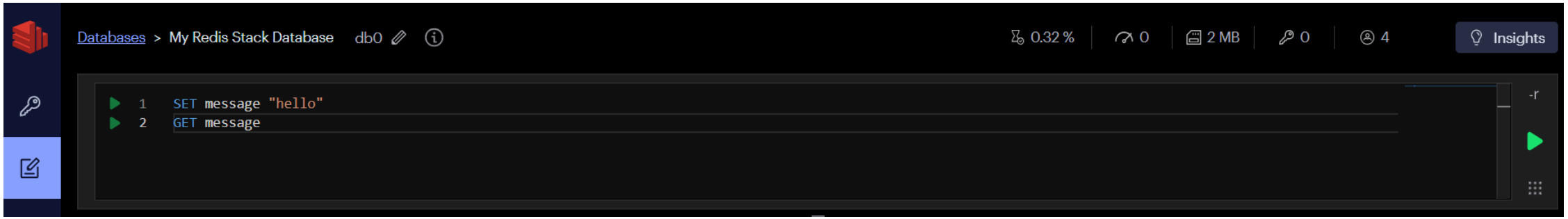
String 활용

1. Redis String

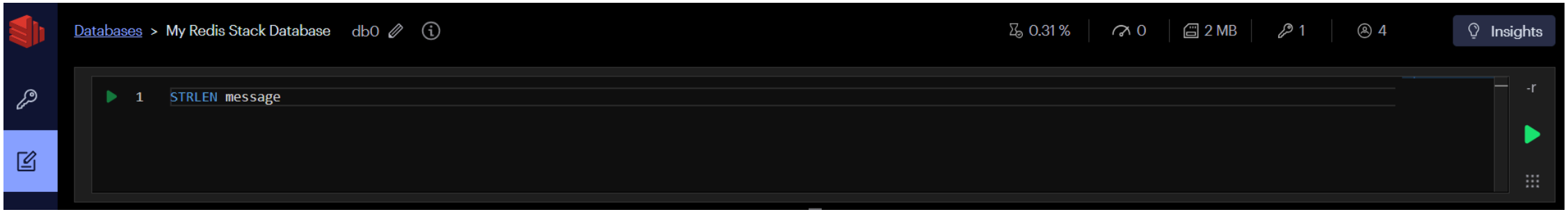
String은 Redis의 가장 기본적인 자료형입니다. 텍스트, 정수, 실수, JSON, 직렬화된 데이터 등을 저장할 수 있습니다. 값의 최대 크기는 크지만, 큰 값을 많이 저장하면 네트워크와 메모리 비용이 커지므로 캐시 대상을 적절히 나누어야 합니다.

```
# message라는 Key에 "hello" 문자열 저장
SET message "hello"

# message의 값 조회
GET message
```

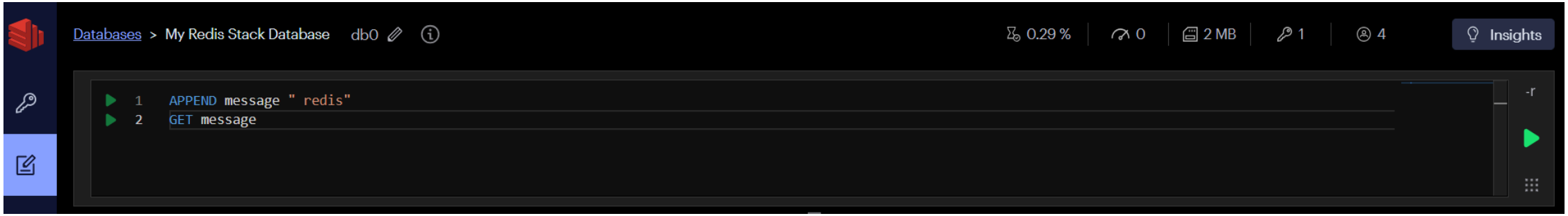


```
# message 문자열의 길이(문자 수) 조회
STRLEN message
```

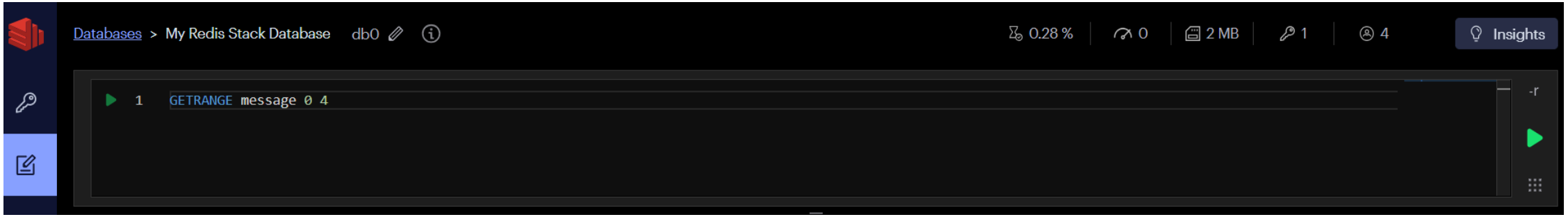


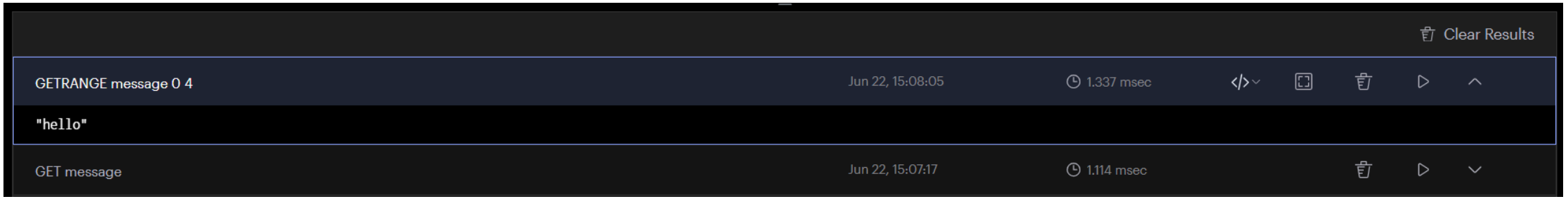
```
# message 문자열 끝에 " redis" 추가
APPEND message " redis"

# 변경된 message 값 조회
GET message
```



```
# message 문자열의 일부만 조회 (0번부터 4번 인덱스까지)
GETRANGE message 0 4
```

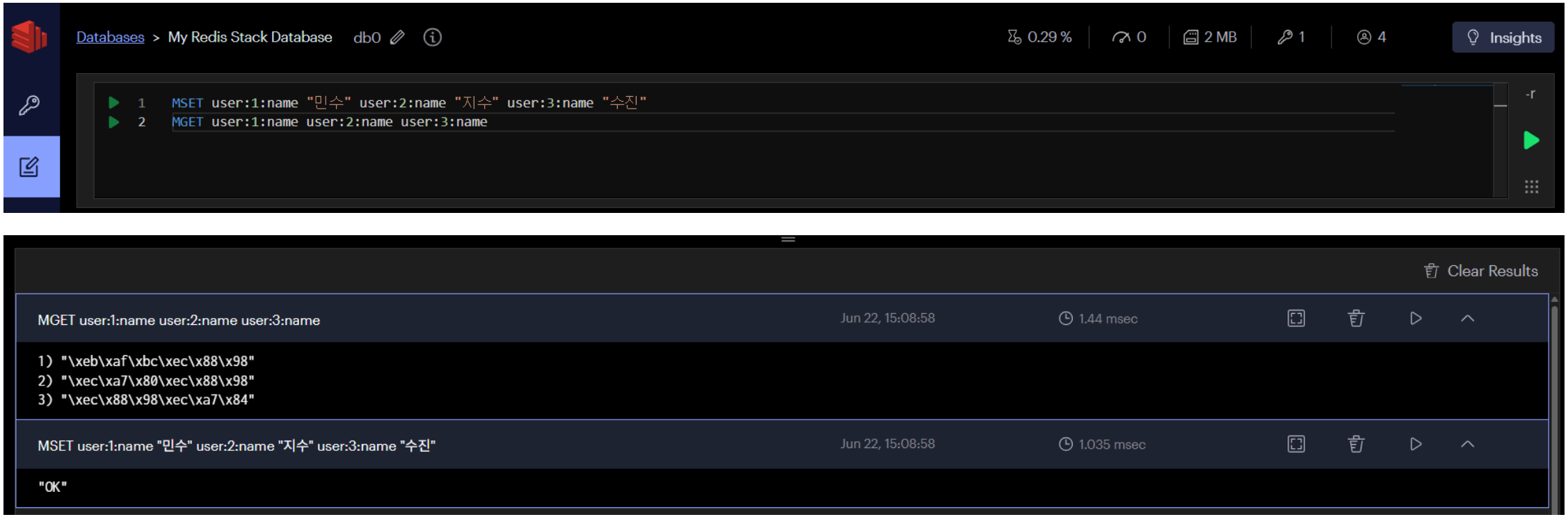




2. 여러 값 처리

```
MSET user:1:name "민수" user:2:name "지수" user:3:name "수진"
MGET user:1:name user:2:name user:3:name
```

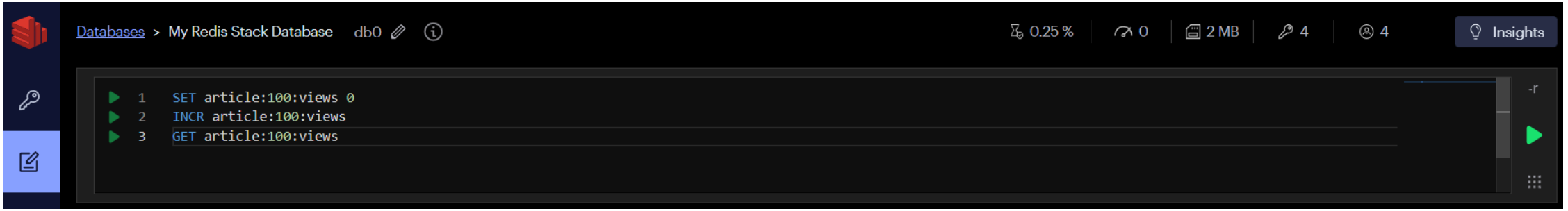
> `MSET`과 `MGET`은 왕복 횟수를 줄일 수 있습니다.
> 다만 서로 관련 없는 많은 키를 무제한으로 한 번에 요청하면 서버와 네트워크에 부담이 되므로 적절한 크기로 나눕니다.


Databases > My Redis Stack Database db0
0.29 %
0
2 MB
1
4
Insights
1 MSET user:1:name "민수" user:2:name "지수" user:3:name "수진"
2 MGET user:1:name user:2:name user:3:name
-r
Clear Results
MGET user:1:name user:2:name user:3:name
Jun 22, 15:08:58
1.44 msec
1) "\xeb\xaf\xbc\xec\x88\x98"
2) "\xec\xa7\x80\xec\x88\x98"
3) "\xec\x88\x98\xec\xa7\x84"
MSET user:1:name "민수" user:2:name "지수" user:3:name "수진"
Jun 22, 15:08:58
1.035 msec
"OK"

3. 카운터

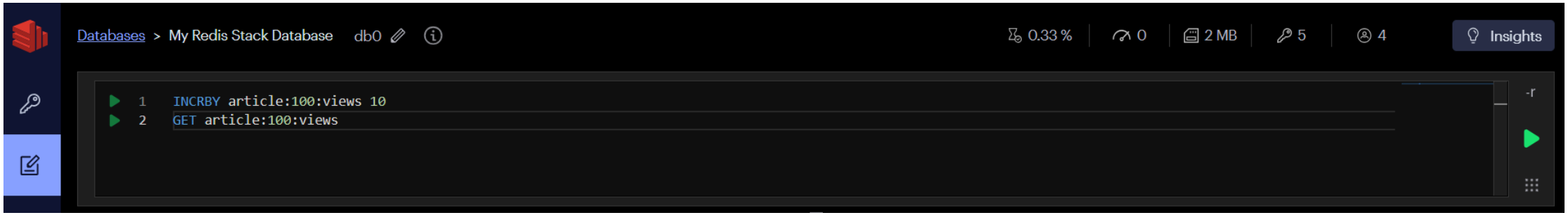
- `INCR` : 값을 1 증가

```
SET article:100:views 0
INCR article:100:views
GET article:100:views
```



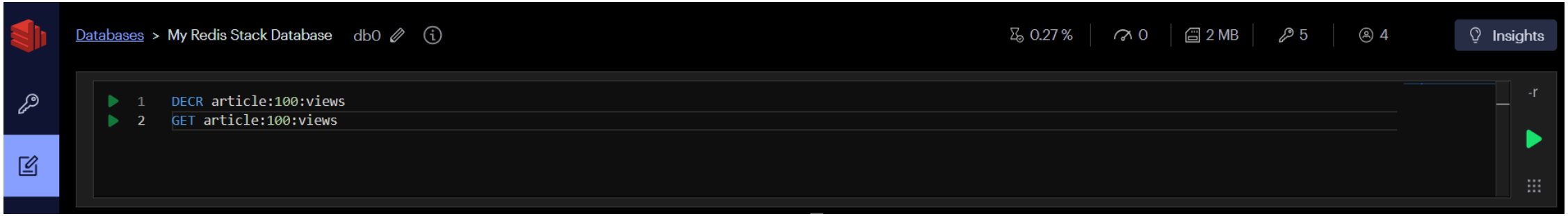
- `INCRBY` : 지정한 정수만큼 증가
- `INCRBYFLOAT` : 실수만큼 증가

```
INCRBY article:100:views 10
GET article:100:views
```



- `DECR` , `DECRBY` : 값을 감소

```
DECR article:100:views
GET article:100:views
```

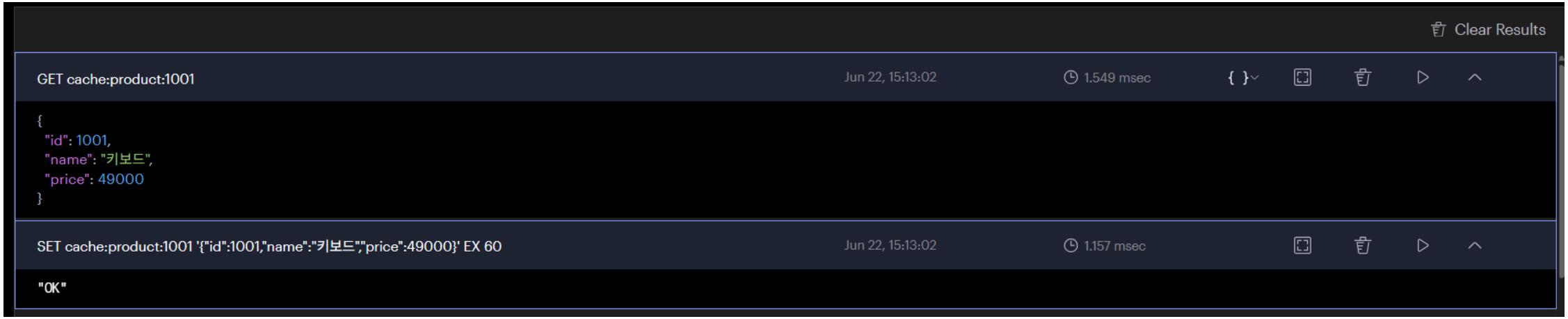


4. JSON 저장

```
SET cache:product:1001 '{"id":1001,"name":"키보드","price":49000}' EX 60
GET cache:product:1001
```

> JSON 문자열은 객체 전체를 한 번에 저장하고 읽기 쉽지만, 일부 필드만 수정하려면 전체 값을 읽어 다시 저장해야 합니다. `필드 단위 조회와 수정이 필요하면 Hash를 고려합니다.`


Databases > My Redis Stack Database db0
0.33 %
0
2 MB
5
4
Insights
1 SET cache:product:1001 '{"id":1001,"name":"키보드","price":49000}' EX 60
2 GET cache:product:1001
-r


Clear Results
GET cache:product:1001
Jun 22, 15:13:02
1.549 msec
{ }
{
"id": 1001,
"name": "키보드",
"price": 49000
}
SET cache:product:1001 '{"id":1001,"name":"키보드","price":49000}' EX 60
Jun 22, 15:13:02
1.157 msec
"OK"